



## examGenome CheatSheet on Algebra

| Sl.no | Content                                                                                                                                                                                                                                                                                                                                      |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | <p><b>What is the <b>mean</b> of a set of numbers?</b></p> <p>Answer: The mean is the average—found by adding all the numbers and dividing by how many numbers there are.</p> <p>Explanation: Example: For 2, 4, 6: add them (<math>2+4+6=12</math>), then divide by 3 <math>\rightarrow</math> mean = 4. It gives a 'fair share' value.</p> |
| 2     | <p><b>What is the <b>median</b> of a list of numbers?</b></p> <p>Answer: The median is the middle number when the numbers are put in order from smallest to largest.</p> <p>Explanation: If there's an even number of values, the median is the average of the two middle numbers. It shows the 'center' of the data.</p>                    |
| 3     | <p><b>What is the <b>mode</b> of a data set?</b></p> <p>Answer: The mode is the number that appears most often.</p> <p>Explanation: A data set can have one mode, more than one mode, or no mode at all (if all numbers appear the same number of times).</p>                                                                                |
| 4     | <p><b>What is the <b>range</b> of a group of numbers?</b></p> <p>Answer: The range is the difference between the largest and smallest numbers.</p> <p>Explanation: You find it by subtracting: Range = Biggest number – Smallest number. It tells how spread out the data is.</p>                                                            |
| 5     | <p><b>Do mean, median, and mode have units?</b></p> <p>Answer: Yes! They use the same units as the data—like centimeters, dollars, or points.</p> <p>Explanation: If you're finding the mean height of students in cm, the answer is in cm. The units don't disappear!</p>                                                                   |
| 6     | <p><b>Which measure of center is not affected by very large or very small numbers (outliers)?</b></p> <p>Answer: The median.</p> <p>Explanation: The median only cares about the middle value, so one very high or low number won't change it much. The mean can be pulled up or down by outliers.</p>                                       |
| 7     | <p><b>Can a data set have more than one mode?</b></p> <p>Answer: Yes! If two (or more) numbers appear the same highest number of times, the data is 'bimodal' or 'multimodal'.</p> <p>Explanation:</p>                                                                                                                                       |
| 8     | <p><b>What must you always do before finding the median?</b></p> <p>Answer: Put the numbers in order from smallest to largest.</p> <p>Explanation: The median is based on position, not value—so ordering is essential. For example: 5, 1, 3 <math>\rightarrow</math> order to 1, 3, 5 <math>\rightarrow</math> median is 3.</p>             |

| <b>Sl.no</b> | <b>Content</b>                                                                                                                                                                                                                                                                                               |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9            | <p><b>Which measure tells you how spread out the data is?</b></p> <p>Answer: The range.</p> <p>Explanation: A small range means numbers are close together. A large range means they are far apart. It's the simplest measure of spread.</p>                                                                 |
| 10           | <p><b>What is the main difference between mean and median?</b></p> <p>Answer: The mean uses all numbers in the calculation, while the median only uses the middle one(s).</p> <p>Explanation: Mean = total ÷ count. Median = middle position. They can be very different if the data has extreme values.</p> |

